SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1*:* Apply AI basics such as intelligent agents and uninformed search methods fo solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Annexure A

Descriptive Written Test

Duration : **45** Minutes **Total Marks:20**

(192411124
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Code of Conduct:

I

1 K. Niharika

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Niharika

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria

Criteria

Weightag

Level 4-
Exemplary

Level 4 (Exemplary)

Level 3
(Proficient)

Demonstrates

Level 1-

Level 3- Proficient

Level **2**
(Developing)

understanding

Level 2-
Developing

Level **1**
(Beginning**)**

Basic
understanding
with some
misconception

Beginnin
g

Max Marks

Poor understandi
ng; major
misconcepti
ons

Understanding of
Concepts

thorough,

Good

25%

accurate

understanding of concepts

with minor gaps

s

Detailed,

well-explained

Clear

Explanation

25%

answers with

adequate

relevant

explanation with

Limited

explanation; lacks depth

Poor or

unclear

explanation

details

examples

Effectively

applies

Applies

Application

20%

concepts with

strong

concepts with minor errors

analytical

ability

Well-structure

Limited

application with weak analysis

d answers with

Organization

15%

logical flow

Organized with

minor issues in flow

Basic structure, lacks clarity

Accuracy 10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	No application or incorrect analysis
Clarity 5%	Clear, neat, professional presentation and coherence	Understandable with minor issues	Limited clarity, readability issues	Poor organization, incoherent answers Incorrect answers with major gaps Poorly written, difficult to understand

bergent Agents in Smart Home Automation

production :

An intelligent agent

B a System that senses its Erveronment and it takes Suttable actions to achieve a goal. In a Smart home, Servers, detect motton, temperature, teme & Security Conditions and the agent Controls lights, fans, Ac & alarms

oo

1. Simple Heflex

Agent ÷

A Simple Heflex agent acts only according to the Current Struation using Condition - action Hules.

Examples ÷

* If motion is detected → turn ON the light

* If Smoke is detected → tum ON the alarm

2. Model Based Agent ÷

It maintains an Internal State of the environment.

It uses both current & previous information.

Ex응

It semembers that a person is already present in a goom & maintains the Status of lights & Temperature.

3. Goal Based Agent

÷

A Goal based agent Selects

Goal

Ex÷

Its

actions to achieve a Specific

be to maintain

a Comfortable temperature

goal may Where Keeping the house Secusie

144. utelety Based Agent:

Introduc

A

based agent Chooses the action that prov

Comf or

the best Overall Sterilt by Cornidering factors such as energy Consumption & Securtly.

Ex ÷

It may use a fan Pontead of AC When the temperature is only Slightly high to Save energy

Best Approach:

For a Smart home

。a

bybred approach using model based, would be most effective

Goal based & utility based agents would be

A Semple Steflex Component can handle immediate actions Such as

Such as detecting

detecting motion (or) Smoke.

A model based Component Can maintain information

about the Current State of the House. A Goal bared Component can work towards goals Such as Comfort & Security. Finally, a utility based Component Can Select the best action by balancing Comfort, energy effectency, Cost & Security.

Conclusion

÷

is

Therefore, a Hybrid intelligent agent System & better than using only one type of agent. It can respond Quickly to Heal time events remember the environment

learn user

Preferences, & make decisions that provede Comfort, energy efficiency & Security

tion that
Such
as
Comfor

Findi

ng

Dduction

a path in an Unknown Maze

A srobot placed in an unknown maze does not know the Complete layout of the enveronment. It must explore different paths untel it finds the exlst.

Since there is Know no preor Knowledge

or

Heurestec

can be used.

Information, unenformed Search Strategles

Towo important unenformed Search methods are:

#uneform Cost Search (ucs)

Iterative Deepening Search (IDS)

Uniform Cost Search [ucs]

÷

*uneform Cost Search expands the node with lowest path cost * It uses preorety queue to Select the cheapest path Working ÷

* Start from the initial posetion

* Generate possible Neighboring posetions

* Calculate the Cost of reaching each posetion

Select the position with the Lowest total path cost * Contenue until the exet is reached.

Ex÷

If different paths have different movemental Costs, UCS Chooses the path With menemum total Cost.

Advantages

÷

** Complete of Step Costs are positive

* optimal because it finds the least Cost Soution

* Useful When different paths have different Costs

Disadvantag es

=

* Can use a large amount of

memory

It takes 18g time of there are many paths avaliable

we

It doesn't use disrection of the Goat.

Iterative Deepening Search (IDS)

Iterative

Deepenin

g

Can Wr

Search Combines the advantages of depth first Search (DFS) & Breadth first Search (BFS)

#It Stepeatedly performs depth limited DFS with increaving depth lemets.

Ex÷

Depth o→ Search

Depth 1

Search

Depth 2 Search

Depth 3

Search

exet

It Continues until the gobot finds the exit.

Advantages

:

* Requires much less memory than BFS

*Complete When the branching factor is fenete.

**

fends the Shallowest Solution When all Step Costs are equal.

vantage

s:

Nepeats the Same nodes

Multiple temes

* Can Waste

Computation

*Not optimal When different actions have different Conte

UCS vs IDS ÷

IDS

Increasing depth limet

yes only for equal steps costs

Feature

UCS

Selection

Lowest path cost

Comple
te

Yes, with poseteve
Costs

Yes

optimal

Yes

Memor
y

Suitabl
e for different
costs

Advantage

ttigh

Yes

Menemum Cost
path

Low

No

Memory

effecent

Conclusion:

Therefore

a normal maze Where Each movement has the Same Costs, a model based & Goal based agent Combined With IDS & an effective Choice because it uses Less memory & can find the exit even When the Solution depth is unknown.